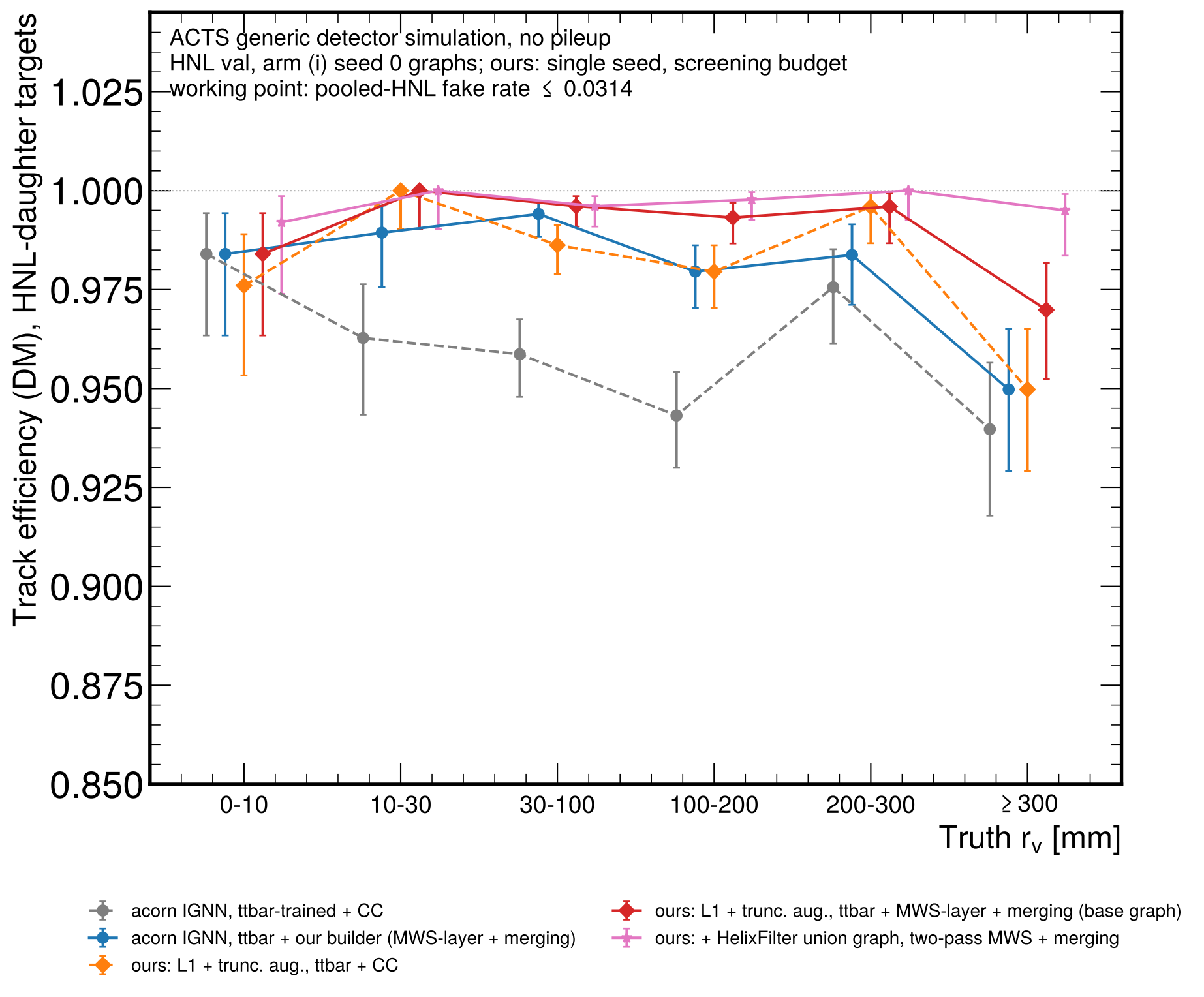
ACTS generic detector simulation, no pileup
HNL val, arm (i) seed 0 graphs; ours: single seed, screening budget
working point: pooled-HNL fake rate ≤ 0.0314
Track efficiency (DM), HNL-daughter targets
1.025
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0-10
10-30
30-100
100-200
200-300
≥ 300
Truth r_v [mm]
acorn IGNN, ttbar-trained + CC
acorn IGNN, ttbar + our builder (MWS-layer + merging)
ours: L1 + trunc. aug., ttbar + CC
ours: L1 + trunc. aug., ttbar + MWS-layer + merging (base graph)
ours: + HelixFilter union graph, two-pass MWS + merging